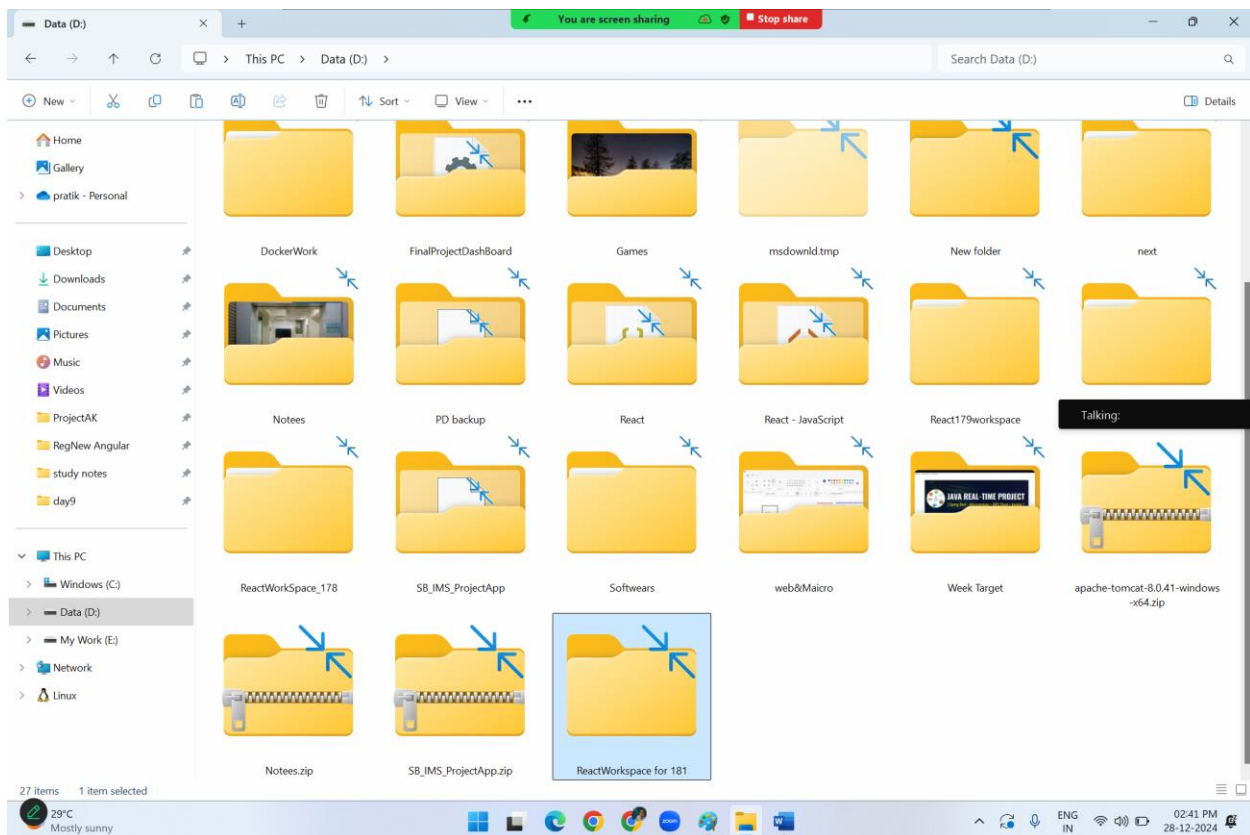




by Kunal Sir

Creating React Application using Vite:

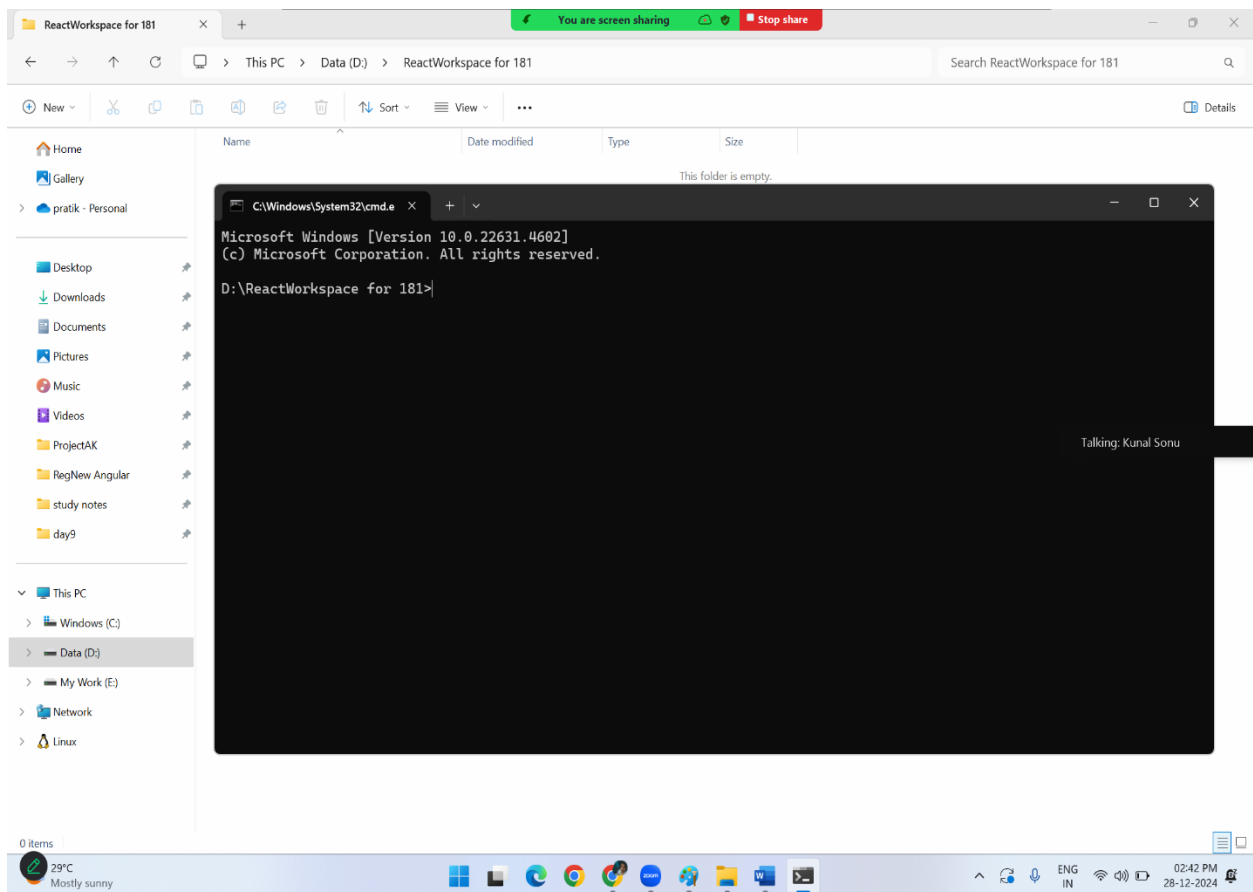
Step 1: - Create A workspace for react projects.





by Kunal Sir

Step 2: - Open workspace directory and launch CMD.

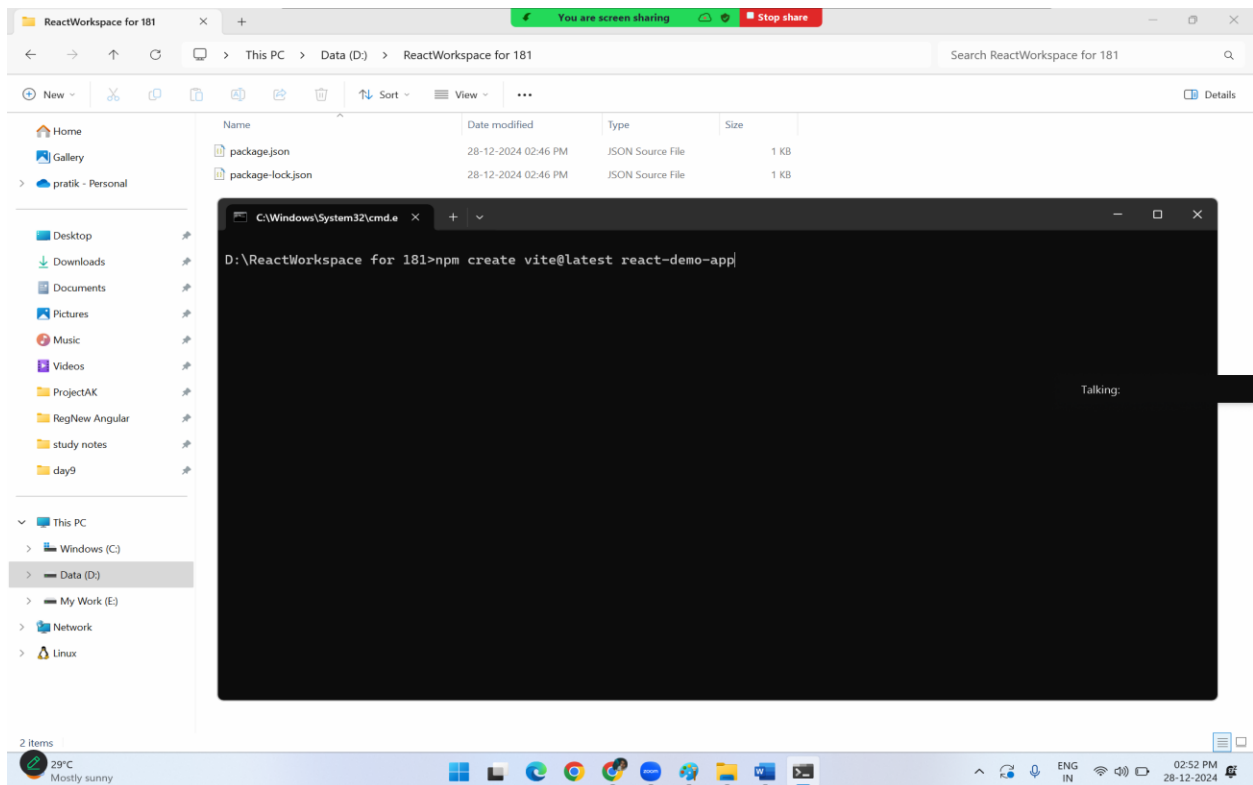


Stop, Near, 1st Floor, Above Rupam Sweets/ Priyanka Collections Building Vikas Mitra Mandal Chowk
Road, Karve Nagar, Pune, Maharashtra 411052 , Mobile No.- 8888022204



by **Kunal Sir**

Step 3: - Create React application using vite tool.



Stop, Near, 1st Floor, Above Rupam Sweets/ Priyanka Collections Building Vikas Mitra Mandal Chowk
Road, Karve Nagar, Pune, Maharashtra 411052 , Mobile No.- 8888022204

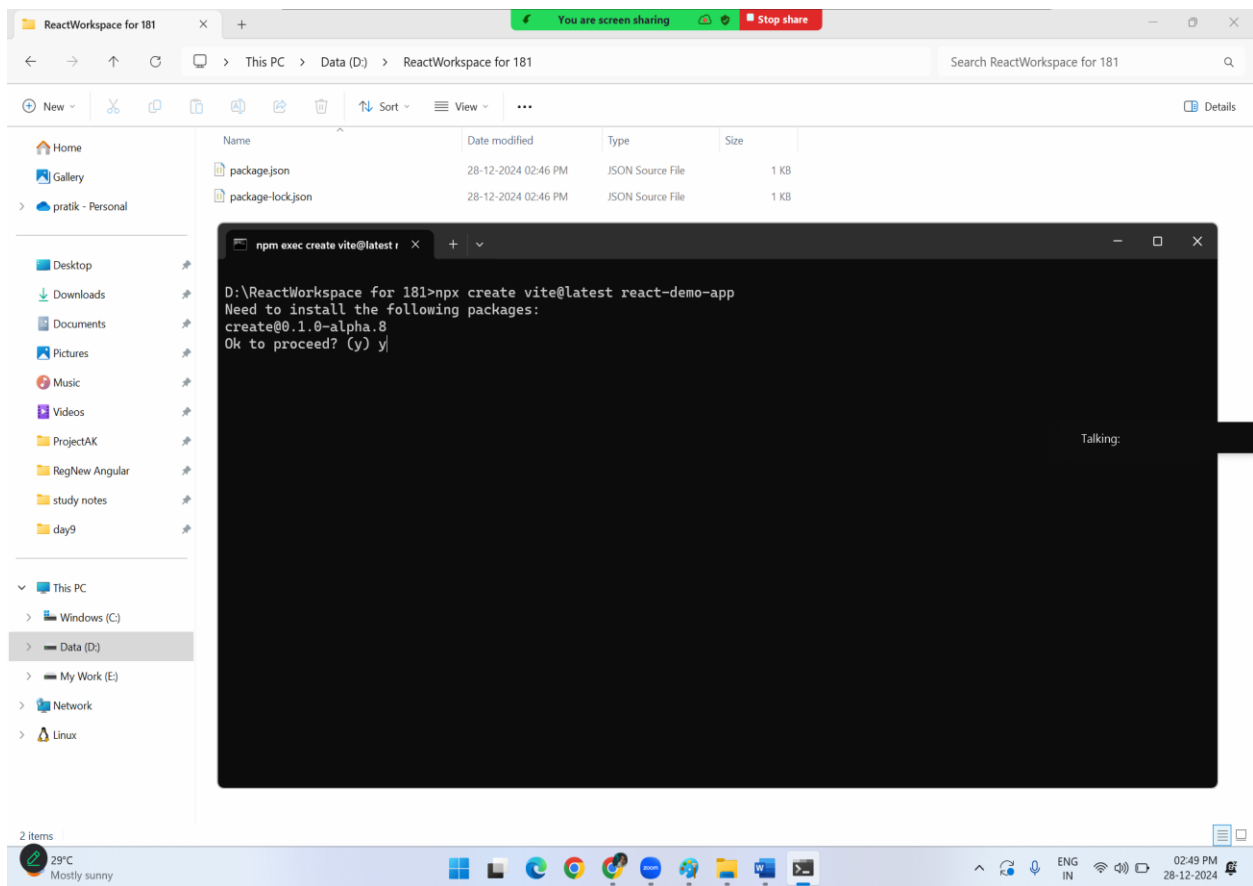


CJC

Complete Java Classes

by **Kunal Sir**

Step 4: - for the first time only, it will ask to install vite tool. Enter `y` to proceed.



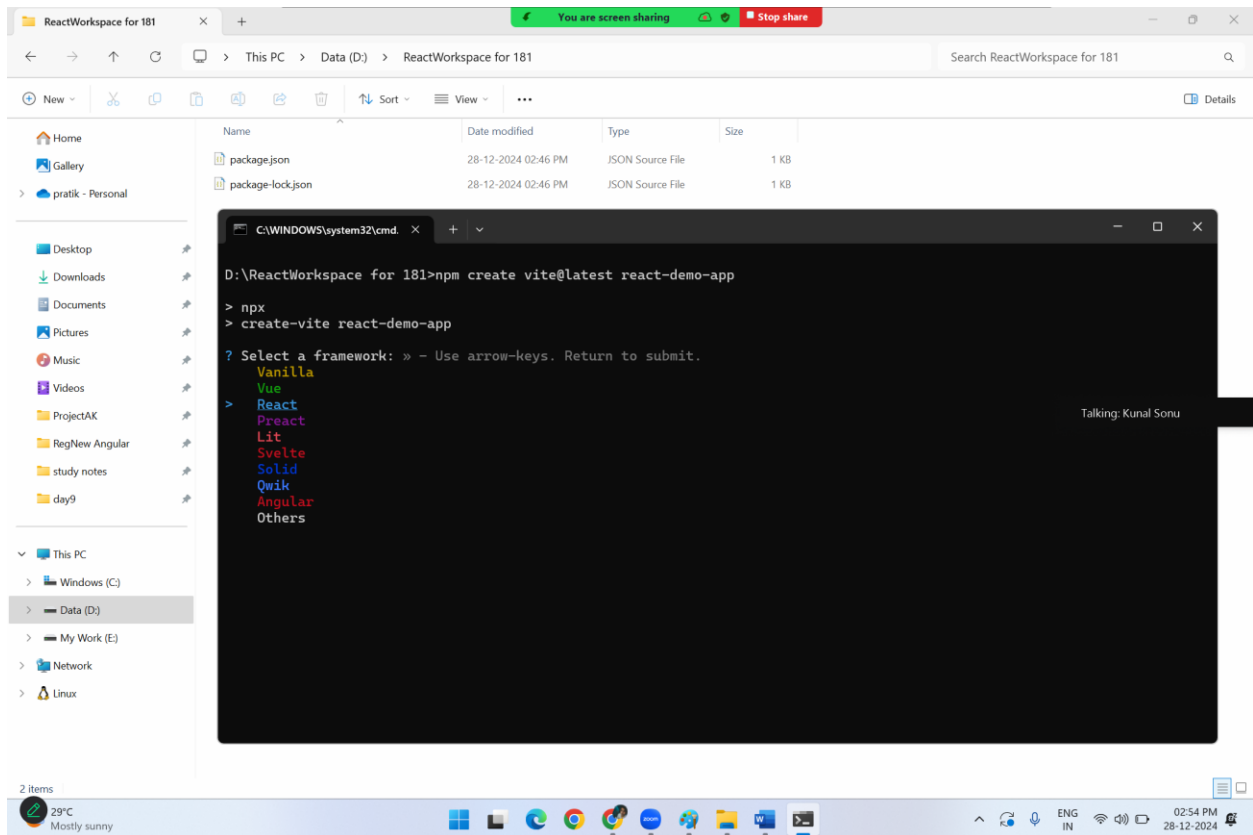
Stop, Near, 1st Floor, Above Rupam Sweets/ Priyanka Collections Building Vikas Mitra Mandal Chowk
Road, Karve Nagar, Pune, Maharashtra 411052 , Mobile No.- 8888022204



by **Kunal Sir**

Step 5: - After Enter, it will install require packages and further proceed to create application, we need to select out project framework/library which we want to use.

We can switch between given option using up-key or down key





CJC

Complete Java Classes

by Kunal Sir

Step 6: Now we have select variant/programming language. Select JavaScript as a programming language.

The screenshot shows a Windows File Explorer window titled "ReactWorkspace for 181" with a search bar and a list of files: "package.json" and "package-lock.json", both dated "28-12-2024 02:46 PM" and "JSON Source File" type, each "1 KB" in size. Overlaid on this is a Command Prompt window titled "C:\WINDOWS\system32\cmd." showing the following commands and output:

```
D:\ReactWorkspace for 181>npm create vite@latest react-demo-app
> npx
> create-vite react-demo-app
? Select a framework: > React
? Select a variant: > - Use arrow-keys. Return to submit.
  TypeScript
  TypeScript + SWC
  > JavaScript
  JavaScript + SWC
  React Router v7 /
```

The taskbar at the bottom shows the system clock as "02:56 PM 28-12-2024" and the weather as "29°C Mostly sunny".

Stop, Near, 1st Floor, Above Rupam Sweets/ Priyanka Collections Building Vikas Mitra Mandal Chowk Road, Karve Nagar, Pune, Maharashtra 411052 , Mobile No.- 8888022204



by **Kunal Sir**

Step 7: - Once we specify programming language, vite will generate the project structure in our workspace directory.

The screenshot shows a Windows File Explorer window titled 'react-demo-app' with a search bar containing 'react-demo-app'. The left sidebar shows the 'Data (D:)' drive. The main pane displays a table of files and folders: 'public' (File folder, 28-12-2024 02:57 PM) and 'src' (File folder, 28-12-2024 02:57 PM). Overlaid on the File Explorer is a Windows Command Prompt window. The command prompt shows the following commands and output:

```
D:\ReactWorkspace for 181>npm create vite@latest react-demo-app
> npx
> create-vite react-demo-app
? Select a framework: > React
? Select a variant: > JavaScript
Scaffolding project in D:\ReactWorkspace for 181\react-demo-app...
Done. Now run:
cd react-demo-app
npm install
npm run dev
D:\ReactWorkspace for 181>
```

The Command Prompt window also shows a 'You are screen sharing' notification at the top and a 'Talking: Kunal Sonu' status bar at the bottom.

Note: - Now, this project will not have any required libraries yet. To get/download the required libraries we need to perform the following steps



CJC

Complete Java Classes

by **Kunal Sir**

Step 8: - move the command prompt directory to the project directory using `cd` command.

The screenshot shows a Windows File Explorer window titled "ReactWorkspace for 181" with the address bar set to "D:\ReactWorkspace for 181". The left sidebar shows the "Data (D:)" drive selected. The main pane displays a folder named "react-demo-app" created on 28-12-2024 at 02:57 PM. Overlaid on this is a Windows Command Prompt window titled "C:\Windows\System32\cmd.exe". The command prompt shows the following commands and output:

```
D:\ReactWorkspace for 181>npm create vite@latest react-demo-app
> npx
> create-vite react-demo-app
? Select a framework: > React
? Select a variant: > JavaScript
Scaffolding project in D:\ReactWorkspace for 181\react-demo-app...
Done. Now run:
  cd react-demo-app
  npm install
  npm run dev

D:\ReactWorkspace for 181>cd react-demo-app
D:\ReactWorkspace for 181\react-demo-app>
```

The taskbar at the bottom shows the system clock as 03:05 PM on 28-12-2024, with the temperature at 29°C and weather as "Mostly sunny".



CJC

Complete Java Classes

by **Kunal Sir**

Step 9: - Now using `npm install` command we can download all the require dependencies which is specified in package.json file in project directory.

The screenshot shows a Windows File Explorer window titled 'ReactWorkspace for 181' with a search bar and navigation pane. The main pane displays a folder named 'react-demo-app' created on 28-12-2024 at 03:09 PM. Overlaid on this is a Command Prompt window titled 'C:\Windows\System32\cmd.exe'. The command prompt shows the following commands and output:

```
D:\ReactWorkspace for 181>npm create vite@latest react-demo-app
> npx
> create-vite react-demo-app
? Select a framework: > React
? Select a variant: > JavaScript
Scaffolding project in D:\ReactWorkspace for 181\react-demo-app...
Done. Now run:

  cd react-demo-app
  npm install
  npm run dev

D:\ReactWorkspace for 181>cd react-demo-app
D:\ReactWorkspace for 181\react-demo-app>npm install
added 254 packages and audited 255 packages in 21s
105 packages are looking for funding
  run 'npm fund' for details
found 0 vulnerabilities
D:\ReactWorkspace for 181\react-demo-app>
```

The Windows taskbar at the bottom shows the system clock as 03:10 PM on 28-12-2024, with a weather widget indicating 29°C and 'Mostly sunny'.



CJC

Complete Java Classes

by Kunal Sir

Step 10: - type `code .` to launch the VS-Code ide into project root directory.

```
D:\ReactWorkspace for 181>npm create vite@latest react-demo-app
> npx
> create-vite react-demo-app
Select a framework: > React
Select a variant: > JavaScript
Scaffolding project in D:\ReactWorkspace for 181\react-demo-app...

Done. Now run:

  cd react-demo-app
  npm install
  npm run dev

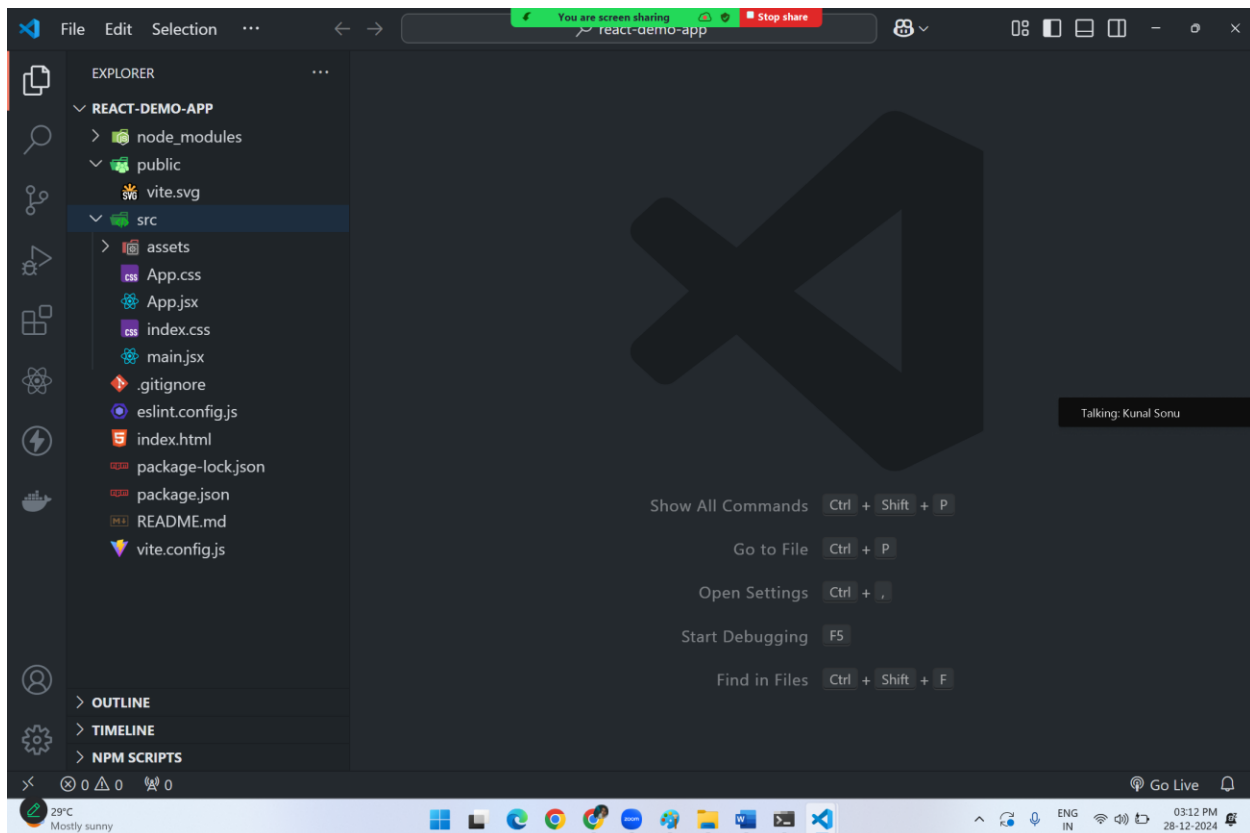
D:\ReactWorkspace for 181>cd react-demo-app
D:\ReactWorkspace for 181\react-demo-app>npm install
added 254 packages, and audited 255 packages in 21s
105 packages are looking for funding
  run 'npm fund' for details
found 0 vulnerabilities
D:\ReactWorkspace for 181\react-demo-app>code .|
```



CJC

Complete Java Classes

by Kunal Sir

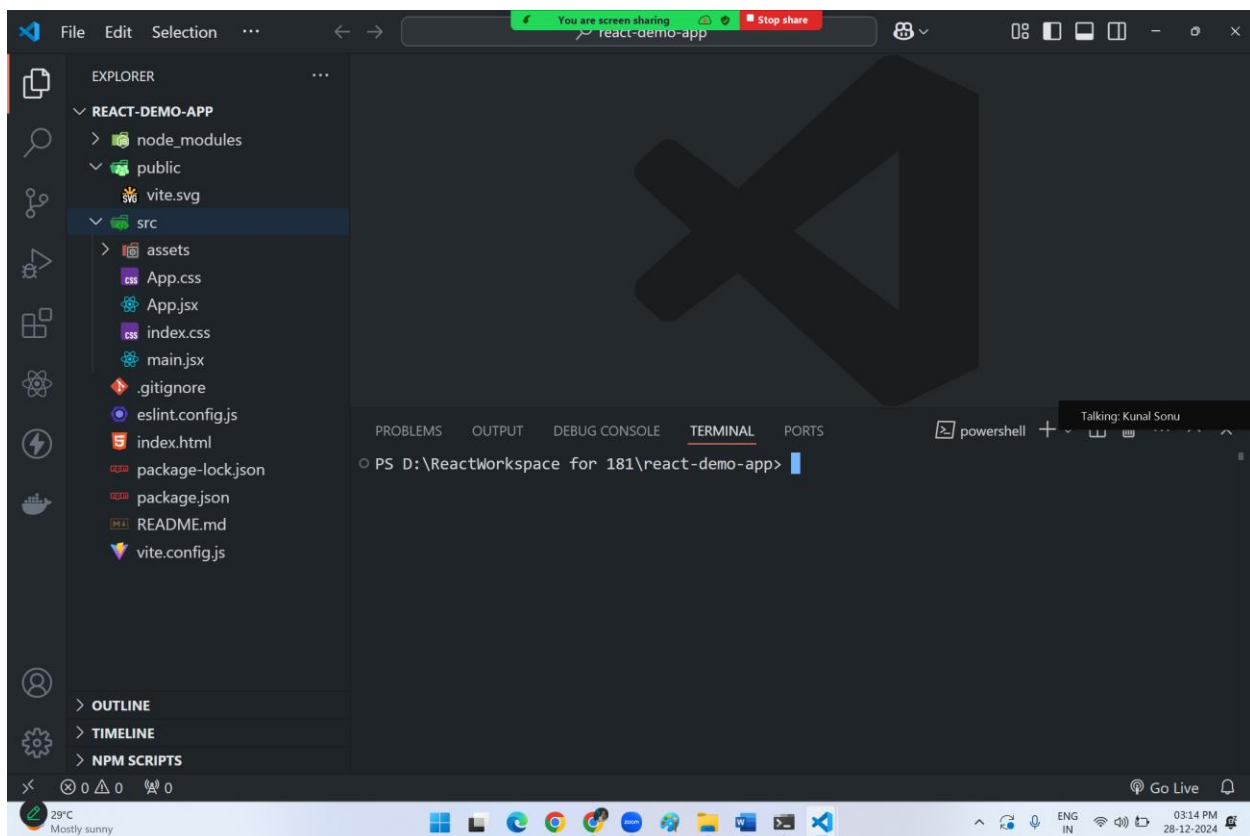


Complete Java Classes



by **Kunal Sir**

Step 11: - open the VS-Code's terminal to run the project.





CJC

Complete Java Classes

by **Kunal Sir**

Step 12: - to run the project enter `npm run dev` command in terminal.

The screenshot shows the Visual Studio Code interface. On the left, the Explorer panel displays the project structure for 'REACT-DEMO-APP', including 'node_modules', 'public', 'src', and various configuration files. The main editor area shows a large 'X' watermark. At the bottom, the Terminal panel is active, showing the command 'npm run dev' being executed. The output indicates that Vite v6.0.6 is ready in 207 ms and provides the local development URL: 'http://localhost:5173/'.

```
PS D:\ReactWorkspace for 181\react-demo-app> npm run dev

> react-demo-app@0.0.0 dev
> vite

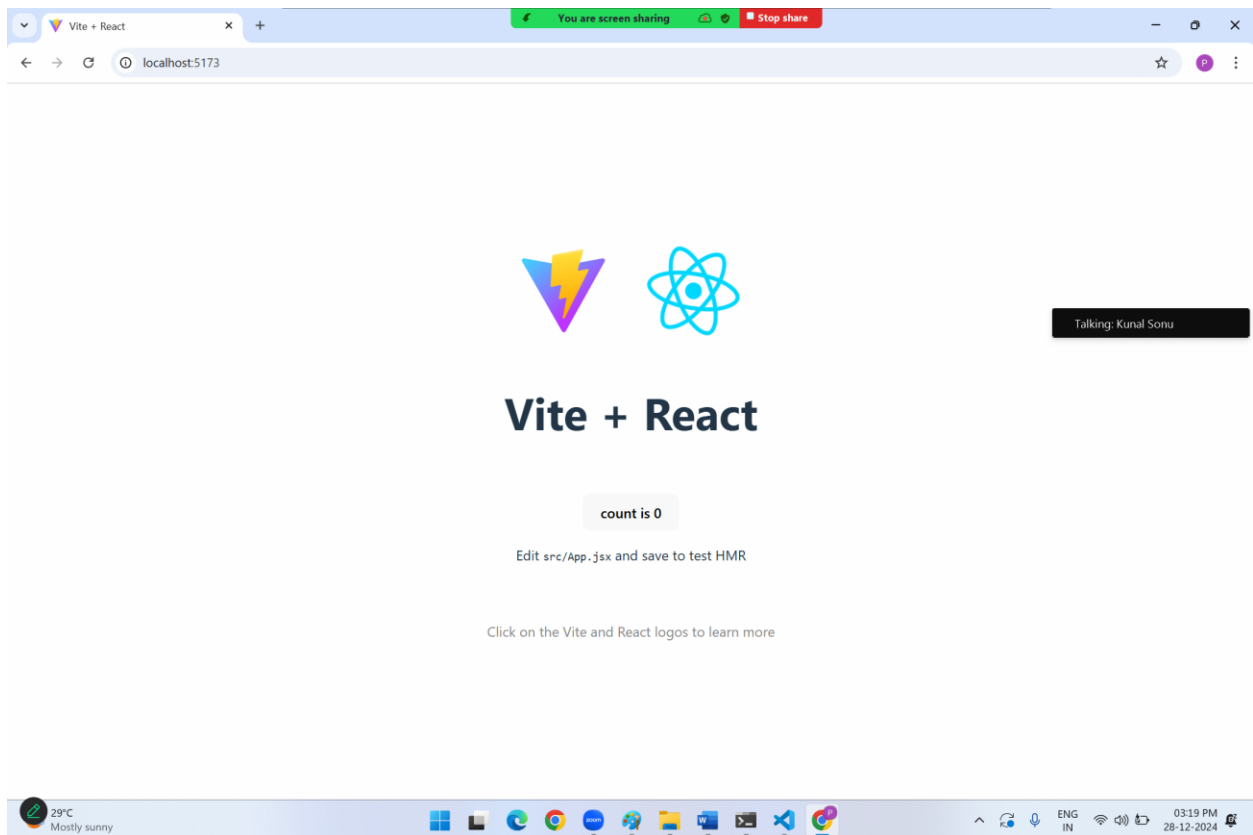
VITE v6.0.6 ready in 207 ms

  → Local:   http://localhost:5173/
  → Network: use --host to expose
  → press h + enter to show help
```



by Kunal Sir

Step 12: - Now, Vite will execute run script to run the react application in dev environment and Vite will launch the react application on it's default port number 5173. We can visit to the application through web browser using url: <http://localhost:5173>.



Stop, Near, 1st Floor, Above Rupam Sweets/ Priyanka Collections Building Vikas Mitra Mandal Chowk
Road, Karve Nagar, Pune, Maharashtra 411052 , Mobile No.- 8888022204



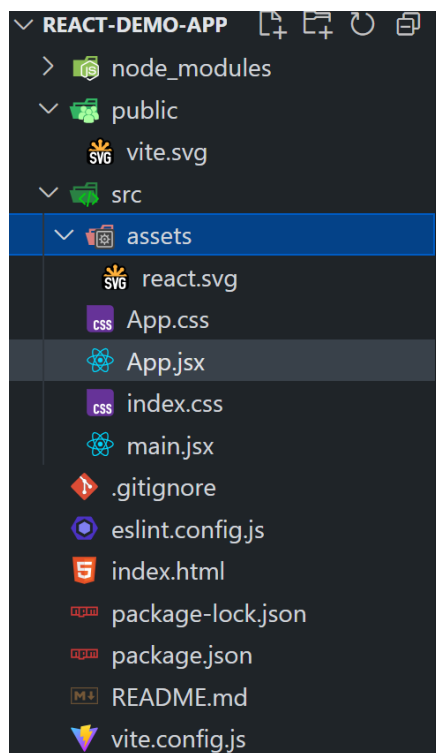
CJC

Complete Java Classes

by Kunal Sir

❖ React Folder Structure: -

When we create the React project using Vite tool, Vite add Project Structure Which includes some of the files and directories which are as below.



CJC

Complete Java Classes

1. **node_modules:** - this directory is generated after we execute `npm install` command. And it contains all the inbuild as well as third party libraries.
2. **Public:** - this directory contains the static content like image, videos ,audios which we may use on our index page
3. **Src :** - this directory includes all the source code of our react application, while developing the react application we will mostly interact with this directory.



CJC

Complete Java Classes

by Kunal Sir

- a) **assets:** - assets directory also contains static files like image, videos and audio files, but these files are used in source code only.
- b) **App.jsx:** it is the root component of our react application.
- c) **App.css:** this file contains the external css for App component.
- d) **Index.css:** it is the global CSS file and used for write common css code which will be applied to all the React components.
- e) **main.jsx:** - main.jsx is the important file in react application, this file gets executed just after index.html, it contains the render method to render App component.
- f) **.gitignore:** this is not react specific file, it is related to version control.
- g) **eslint.config.js:** - this file concerns to the code review and standards.
- h) **Index.html** it is the only html page of our SPA.
- i) **package.json :** - this file contains the project information and dependencies.
- j) **package-lock.json:-** this file contains the information about nested dependencies.